THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Power is the: 1

(a) energy **(b)** work done
(c) force **(d)** rate of doing work

2. The quality or timbre of a sound enables us to: 1

(a) distinguish two sounds of the same pitch and loudness
(b) find its wavelength
(c) measure its speed
(d) find its loudness

3. A ball falls from rest for 2 s. Its velocity (g = 10 m/s squared) is: 1

(a) 5 m/s **(b)** 40 m/s **(c)** 10 m/s **(d)** 20 m/s

4. A fielder draws his hands back while catching a fast ball to: 1

(a) increase the time of stopping and reduce the force
(b) reduce the mass
(c) stop the ball faster
(d) increase the force

5. A lever is a rigid bar that rotates about a fixed point called the: 1

(a) effort **(b)** load **(c)** axle **(d)** fulcrum

6. SONAR stands for: 1

(a) ship navigation and radar **(b)** sound navigation and ranging
(c) sound network and range **(d)** sound night and ranging

7. Speed is a: 1

(a) unit of length **(b)** vector quantity **(c)** force **(d)** scalar quantity

8. 1 newton is the force that gives an acceleration of 1 m/s squared to a mass of: 1

(a) 100 g **(b)** 1 kg **(c)** 10 kg **(d)** 1 g

9. A force of 10 N moves a body through 5 m in the direction of the force. The work done is: 1

(a) 15 J **(b)** 5 J **(c)** 2 J **(d)** 50 J

10. A region in a sound wave where the particles are crowded together is a: 1

(a) rarefaction **(b)** trough **(c)** compression **(d)** crest

11. A straight line parallel to the time axis on a velocity-time graph indicates: 1

(a) retardation **(b)** uniform velocity
(c) increasing acceleration **(d)** rest only

12. Friction always acts: 1

(a) along the direction of motion
(b) at right angles to motion
(c) downward
(d) opposite to the direction of motion or tendency of motion

13. The SI unit of power is the: 1

(a) joule **(b)** watt **(c)** newton **(d)** kg

14. The audible range of frequencies for humans is about: 1

(a) 200 Hz to 2000 Hz **(b)** 20 kHz to 200 kHz
(c) 20 Hz to 20000 Hz **(d)** 2 Hz to 200 Hz

15. A ball is thrown vertically upward. At the highest point its velocity is: 1

(a) zero **(b)** 10 m/s **(c)** maximum **(d)** equal to initial

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is the use of a fixed pulley? 2

17. State the audible range of frequencies of humans. What are infrasonic and ultrasonic sounds? 2

18. Differentiate between speed and velocity. 2

19. State Newton's second law and write the formula. 2

20. When is the work done negative? Give an example. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Derive the expression for the kinetic energy of a body of mass m moving with velocity v. 3

22. Find the wavelengths in air (v = 344 m/s) for sound of frequencies 20 Hz and 20 kHz. 3

23. A train accelerates from 10 m/s to 30 m/s in 10 s. Find the acceleration and the distance travelled. 3

24. Explain why the horse pulling a cart can make the cart move even though the cart pulls the horse back with an equal force. 3

OR

A person raises a load of 30 cm height using a 40 cm wide ramp. Find the length of the ramp and the mechanical advantage. 3

25. State three characteristics of a musical sound and the factor on which each depends. 3

OR

A body is thrown upwards with a velocity of 20 m/s. Find the maximum height and the time to reach it (g = 10 m/s squared). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A car starting from rest acquires a speed of 30 m/s in 6 s and then moves with this speed for 10 s, after which it brakes uniformly to rest in 5 s. Find the acceleration in each stage and the total distance. **5**

OR

Describe an activity to show the inertia of rest and of motion with a coin, a card and a glass or a pile of coins. **5**

27. Derive the work-energy theorem and use it to find the speed of a 2 kg body at rest after a net work of 100 J is done. **5**

OR

Derive v = f lambda and use it to find the speed of sound in air if a wave has f = 256 Hz and lambda = 1.34 m. **5**

END OF QUESTION PAPER — Total: 50 Marks

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